Aviation's Public Health Footprint in New England

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Introduction

Aviation in New England exists in an unusual position. It is essential economic and transportation infrastructure, but can cause meaningful, measurable harms. While much of the benefit and harm comes from the everyday activities of aircraft and airports, there are some more extreme effects, both positive and negative. Compare Brunswick's 2024 fire-fighting foam spill that released tens of thousands of gallons of PFAS into the ecosystem with needing emergency transportation to Boston on LifeFlight of Maine's "Lima Mike." Neither are everyday occurrences, but both can have massive impacts on the lives of Mainers.

Most aviation health research focuses on the busiest airports in the largest cities, while rural and remote areas remain understudied. Maine is a perfect example of this phenomenon; its commercial airports are small and its municipal airports are even smaller. With only one interstate highway serving Maine, the only access to time-sensitive transportation for rural communities like those in Piscataquis county or the islands in Penobscot Bay is by air. Recognizing this, the Federal Government subsidizes four Essential Air Service routes in Maine, Augusta, Bar Harbor, Presque Isle, and Rockland, placing it in the top ten nationally, and first in New England.

The primary focus of this paper is to answer the question of "To what extent does aviation activity contribute to adverse health outcomes and how does this compare to what broader research says about noise, ultrafine particles, lead, and PFAS?" The approach will focus on data sets consisting of enplanement activity, modeled cancer risk, and pollutant exposure At its core, aviation policy is public health policy, inseparable from economic policy. Those most vulnerable to aviation's harms, especially those in larger cities near busier airports, do not have the political capital to enact change needed for just outcomes. Policy is often decided without regard for who it will impact.

This paper will consider the following topics and risk factors. First, it will provide a background on aviation activity in New England and sources for public health concern. The following section reviews literature related to the four primary risk exposure pathways from aviation. These include contamination from airport chemicals (PFAS and deicing fluid), ultrafine particulate emissions from turbine aircraft, lead emissions from piston general aviation, and noise from arriving and departing aircraft, concluding with a discussion of risks and benefits associated with Maine's aviation industry and a connection to the broader New England region.

Background

Aviation in New England operates in a tiered structure. Atop the hierarchy is Boston's Logan airport which handled over 21 million passengers in 2019, more than the rest of the region combined. Manchester-Boston Regional, Bangor International, Portland International Jetport, Providence's T.F. Green, and Burlington International comprise the second tier. The third tier see far less traffic than any of these, seeing mostly general aviation and the occasional charter flight. There are well over one hundred of these public use airports. Maine alone has more than 30, though only Portland and Bangor have air traffic control towers. As a result, accurate modeling becomes challenging due to the non-linear relationship of enplanements and health risk. An airport without the infrastructure for even comparatively small, low impact charter operations may have zero enplanements but could still have a small flight school with two Cessna training aircraft generating meaningful amounts of exposure.

The ground component of airport operations involves a baseline level of activity, regardless of size. Fueling requires the storage and distribution of many thousands of gallons of Jet-A and leaded aviation gasoline, with associated spill risks. Winter operations require the use of glycol-based de- and anti-icing fluid which, despite improvements in collection

technology, still have negative impacts on surrounding watersheds. These activities are not unique to large airports however. Brunswick Executive airport, former home of a naval base and current home of an aircraft recycling facility, has never had scheduled airline traffic and a faulty fire suppression system caused the sixth largest PFAS spill since 1996.

The airborne component of pollution produces a different set of exposures than ground operations. Turbine aircraft emit ultrafine particulate matter (UFP), nitrogen oxides, and other harmful byproducts. Concentrations are highest near the extended centerlines of runways, though few airports see enough traffic for this to be of significant concern. Piston aircraft, which include general aviation aircraft as well as those providing Essential Air Service to Maine, burn leaded gasoline which can deposit trace amounts of lead in the soil and air near areas of operation, disproportionately impacting communities near small airports. Aircraft noise can also impact communities through peak traffic times in addition to health risks from disruption of sleep and increased stress levels. The FAA's NextGen airspace modernization program concentrated flight paths, reducing the total population affected but increasing exposure among those in affected areas.

The heavily regulated environment in which aviation operates also contributes to the present day situation. The FAA does not face much direct pressure from citizens and regulates internally without direct input from elected officials. It is also notoriously slow moving, especially with regard to general aviation which lacks the funding and financial incentive to improve. While piston aircraft are all but gone in commercial service today, they make up the majority of the general aviation fleet. The extremely expensive and lengthy process to certify new aircraft, coupled with the lack of willingness to spend that exists for airliners, results in many aircraft using designs and technologies from the 1940s. The Beechcraft Bonanza which recently ended its run as the longest continuously produced aircraft in history first flew in 1945. While most of these often antiquated technologies do

not impact the non-flying public, the FAA's refusal to approve a successor to 100LL aviation gasoline means pilots and non-pilots alike can experience unnecessary exposure to lead. Though the health risks can be a meaningful localized concern, aviation does have a positive impact on public health in Maine. Organizations such as LifeFlight of Maine, an air ambulance service, offer critical transport for patients needing increased levels of care, while other programs such as the Essential Air Service are continued, in part, due to their ability to expand more routine access to specialized healthcare for residents of more rural and remote areas. Given the multitude of positive and negative impacts of aviation on public health in Maine, it is important to ensure legislation implemented is carefully written in order to maximize aviation's positive impact while minimizing its harms. The following literature review will first examine the impacts of contamination from airport ground operations, then health effects from noise and engine emissions, and conclude with aviation's role as essential infrastructure.

Literature Review

Many of aviation's long-lasting environmental impacts are due to the concentrated ground areas in which they operate. The inherent limitations of fixed-wing aircraft require the concentration of ground operations to the ramp[1] and airport grounds. Aircraft operations require the use and disposal of hazardous substances including aircraft fuel (typically Jet A-1[2] and Aviation Gasoline[3]), firefighting foam[4], deicing fluid, maintenance chemicals, and other waste products. Due to the concentrated nature of the impact of ground operations, much of the public health focus in rural areas will revolve around these sources.

[1]The area of an airport where aircraft are parked, unloaded or loaded, refueled, boarded, or maintained
[2]A highly-refined kerosene-grade fuel similar to diesel
[3]Typically 100LL, a high-octane, ethanol free gasoline with added lead for performance and reliability
[4]Often containing PFAS

Some of the most serious contamination risks are per- and polyfluoroalkyl substances (PFAS). Aeppli et al. focuses on the aftermath of the airport firefighting foam spill at the former Naval Air Station in Brunswick, Maine. This paper focuses on the aquatic ecosystem impact of the PFAS-based foam used in the inadvertently activated hangar fire suppressant system. This research focuses primarily on the spread of the chemicals through waterways in Maine but also contains information on the potential environmental impacts of affected areas through tap and ground water analysis. The Feasibility Assessment for Epidemiological Studies at Pease International Tradeport was written to determine if a study researching the health impact of contaminants present at Pease International Airport might produce meaningful results. Though not in Maine, Pease Air Force Base shares many similarities to the one at Bangor International Airport and the data contained within could be used as an effective proxy. Most of the research looks at the impact of PFAS exposure to children and adult cancers and other diseases for those with more acute exposure. The Brunswick and Pease sources combine to show the initial causes, environmental pathways and health endpoints of PFAS contamination. Outside accidental contamination, routine airport operations may also cause environmental contamination. The EPA's Environmental Impact and Benefit Assessment focuses on the environmental impact of aircraft deicing as well as improvements expected from future regulation changes. This report contains a significant quantity of information ranging from the impacts of deicing fluid on short and long term human health, effects on local aquatic life, as well as concerns regarding dyes used and "aesthetically-impaired water." The Waterville Regional Airport Interim Report on the progress of the current "master plan" contains data concerning a multitude of potential economic and health effects the airport has on the community.

Aircraft operations affect communities in close proximity to airports through two distinct but intertwined manners, noise and air pollution. Research studying the health impact

of both has increased considerably in recent years. The two scientific papers used to study the impact of aircraft noise on public health come from Peters et al. and Argys et al. The first study takes evidence from larger commercial airports in the United States and finds replicable association between aircraft noise and adverse cardiovascular outcomes, identifying sleep disturbances as the primary link between noise and heart disease. The second study by Argys et al. seeks to establish causal evidence by using data from the Federal Aviation Administration's (FAA) recently developed NextGen flight path solution. This policy was supposed to provide broad-reaching improvements to safety, traffic management, collision avoidance and environmental impact. While some of these goals were met, Argys found that NextGen unintentionally concentrated aircraft approaches and departures into narrower corridors, reducing the number of people exposed to excessive noise, but increasing the intensity for others. The study found a statistically significant 1.6% increase in low birth weight among mothers exposed to noise, controlling for age, sex, BMI, alcohol consumption, exercise, smoking, and salt intake. Also found were negative cardiovascular outcomes similar to those found in Peters et al. These studies complement each other, finding similar results despite using differing research methodology. Community members near Logan have also reported noticing these effects, though these accounts likely reflect selection bias.

Other sources in the air pollution segment focus on the direct physical byproducts of aircraft activity such as the emission of UFPs and lead from aviation gasoline. Air pollution research can be broken up into two subcategories, ultrafine particle emissions from turbine powered aircraft typically found at larger airports, and lead emissions from smaller piston aircraft at smaller airports. Hudda et al. found that UFP levels, which contribute to a multitude of health complications such as asthma and COPD, increased by twofold at 4 km when wind blew from the airport compared to baseline. UFP also showed signs of penetrating through some HVAC systems and into residential homes. Riley et al. used mobile

monitoring stations in the flight paths for Los Angeles International (LAX) and Atlanta's Hartsfield-Jackson (ATL) airports. Their data show 3-5 fold increases in UFP concentrations in areas under landing approach paths and suggest particle size can distinguish aircraft emissions from automobile emissions which is an important detail, as their effects are often confounded.

Miranda et al. focuses on general aviation's[5] impact through lead emissions. General aviation aircraft typically fly lower than airliners and will often stay in the same location for an extended time[6], potentially increasing pollution concentration in affected areas. This paper finds a strong geospatial association between proximity to airports with leaded avgas use and increased childhood blood levels. A similar source is the EPA's Data and Analysis of Piston-engine Aircraft Emissions of Lead at U.S. Airports which supports the data found in Miranda et al. This is particularly relevant at smaller airports in Maine with flight schools which are present at most public use airports[7]. Given the outsized relative impact of general aviation in Maine, it is important to understand the impacts are not limited to busy commercial airports but also exist in less densely populated areas.

Unfortunately, much of the research on this subject is concentrated around the largest airports in the country, making it challenging to generalize the information presented here. Even Waterville, which is less developed economically compared to a city such as Boston, releases airport Master Plans that discuss the impact of their fuel farms and deicing operations on surrounding neighborhoods. Given Waterville's smaller population and relative economic stagnation, the lack of data concerning airports of this size raises environmental justice questions not well answered by existing literature.

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[5]The International Civil Aviation Organization defines general aviation as "all civil aviation aircraft operations except for commercial air transport or aerial work…"

[6]A major part of flight training is "pattern work", which consists of flying circuits around the airport. This is an important part of traffic control and maximizes takeoffs and landings per flight hour.

[7]Including but not limited to Waterville, Augusta, Lewiston, Brunswick, Bangor, Portland, and Pittsfield

Benefits of Aviation

Though the primary focus of this paper is the negative impact of aviation on public health, it is important to note its positive economic and public health impact. One of the most visible ways in which aviation benefits Mainers are the services provided by LifeFlight of Maine. According to their annual report, in 2025 LifeFlight transported over 3,000 people, providing them with expedited access to medical care, undoubtedly saving many lives in the process. The Essential Air Service[8] (EAS) is another program with public health benefits. A paper studying these effects by Grubesic & Wei outlines the positive economic impact of EAS on local economies as well as expanding access to nonemergency medical care for residents of rural areas. While the benefits of aviation are not as immediately obvious as those of more remote states such as Alaska, any action taken to mitigate harm to public health from aviation must ensure it does not excessively restrict the critical infrastructure many Maine residents rely on.

Aviation operations cause well-researched and documented environmental and health costs. These include PFAS contamination from firefighting foam, UFP exposure from jet exhaust, noise-induced cardiovascular risk, and adverse birth outcomes. Despite these concerns, aviation also provides rural communities with important services for which there is no comparable substitute. There is a striking asymmetry between the evidence bases for the benefits as compared to the harms. Research on harms and mitigation strategies at large airports is significant and continuously expanding, while little research has been conducted on rural airports. Health impacts at smaller, often less-regulated airports remain relatively unstudied, posing potential environmental justice concerns. While the concerns from UFP and noise are unlikely to abate due to the slow-moving nature of the FAA and lack of suit-

[8] A government program subsidizing air travel to airports that would otherwise not have commercial service.

able replacement technology for aviation turbine engines, ground operations are seeing a revolution in sustainability. The EPA's 2012 deicing assessment established guidelines for controlling the byproducts of aircraft deicing, and the Portland International Jetport was the first airport in the world to use 100% recycled deicing fluid. On a global scale, aviation contributes to climate change via CO2 emission and contrail production, and technology to mitigate their impacts is unlikely to become available in the near future. Overall, data and subsequent action taken from evidence in new research show positive trends in the impact of aviation on public health, but there are still many improvements to be made.

Data and Results

Despite the meaningful amount of research on the impact of aviation on public health, little data analysis exists concerning the regional and general aviation airports that make up the majority of New England's airports. Taking this into consideration, analysis will focus on both busier, commercial airports as well as quieter general aviation airports.

The data contains information from a few related but diverse datasets. First is the EPA's Air Toxics Screening Assessment (AirToxScreen) information on concentrations of potentially harmful chemicals as well as modeled cancer risk estimates due to a wide range of potential factors. This dataset has been joined with the FAAs enplanement data to add the number of airports and enplanements to the list of predictor variables. The AirToxScreen data reflect emissions in the study period ending in 2020, and the enplanement data is from 2019. The primary unit of observation is individual counties in New England. AirToxScreen data was downloaded from the EPA website as Excel files and enplanement data was downloaded from the FAA's Passenger Boarding Data page. My primary outcome variable is modeled cancer risk per million, both overall, and airport-related. Key explanatory variables are airport activity (number of airports and enplanements) and chemical pollution attributable to aviation. AirToxScreen cancer risk is not directly measured, but rather mod-

eled by combining national emissions data with atmospheric dispersion modeling and EPA risk data by pollutant. Limitations of the data include a lack of direct observation of health outcomes and granular flight path data. EPA AirToxScreen data was restricted to the six New England States to match the scope of my research question. Counties with missing population data or cancer risk were filtered. Counties without zero reported enplanement were still kept if they had a public-use airport. In order to mitigate the distortions in data from the COVID-19 pandemic, I used 2019 enplanement data despite EPA data collection concluding in 2020.

Though the intended focus included mostly data from airports in Maine, air travel in this state is not nearly as frequent or highly researched compared to neighboring Massachusetts, home to Boston Logan International Airport, the busiest airport in New England and 16th busiest in the United States by passenger count. While emissions and secondary health risks are undoubtedly a concern for those living around airports, the modeled cancer risk is not extraordinary, even for residents of counties with large, busy airports. Aviation's share of cancer risk is dwarfed by on-road vehicle emissions in every state (Figure 1). The percentage of modeled cancer risk attributable to aviation rarely exceeds 0.5% and in only one case (Nantucket) exceeds 1% (Figure 4). However, airport activity does have a statistically significant connection with modeled cancer risk as shown in Figure 2, and lead emissions from piston aircraft show a similar positive association (Figure 5). Full regression results and diagnostic plots are reported in the Appendix (Table 2; Figure 7). In the bivariate model, a 10% increase in enplanements is associated with a 1.4% increase in modeled airport cancer risk, a relationship that holds across all three models. The number of airports in a county is not statistically significant in any model and its coefficient flips sign between models, suggesting frequency of airport activity matters more than the number of airports.

Table 1: Summary Statistics by State, New England Counties (2020)

State	Pop (M)	Mean Risk	SD Risk	Mean % Airport	SD % Airport	Enpl. (M)
MA	7.03	0.09	0.10	0.54	1.01	21.04
RI	1.10	0.07	0.02	0.31	0.14	2.01
NH	1.38	0.04	0.04	0.16	0.14	0.98
CT	3.61	0.04	0.01	0.15	0.05	3.37
ME	1.36	0.03	0.03	0.15	0.13	1.46
VT	0.64	0.01	0.02	0.06	0.07	0.69

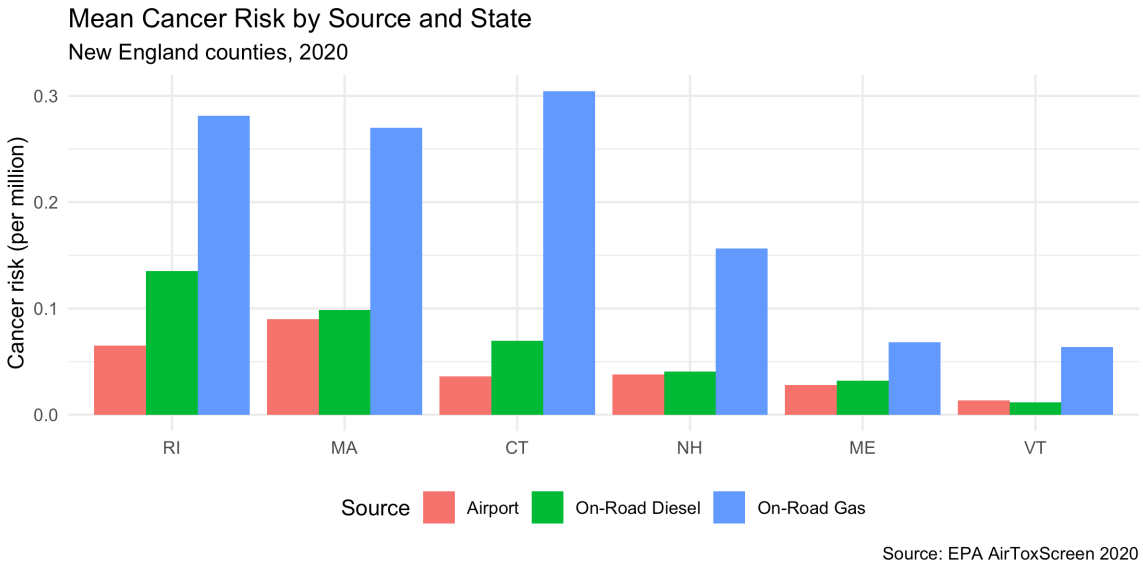


Figure 1: Mean Cancer Risk by Source and State

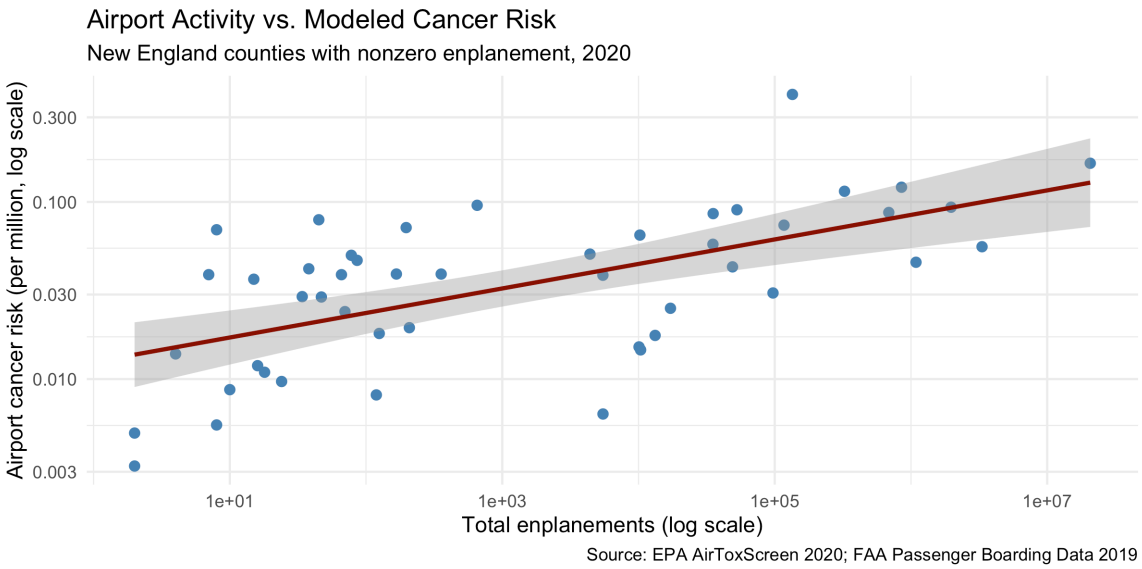


Figure 2: Airport Activity vs. Modeled Aviation-Attributable Cancer Risk

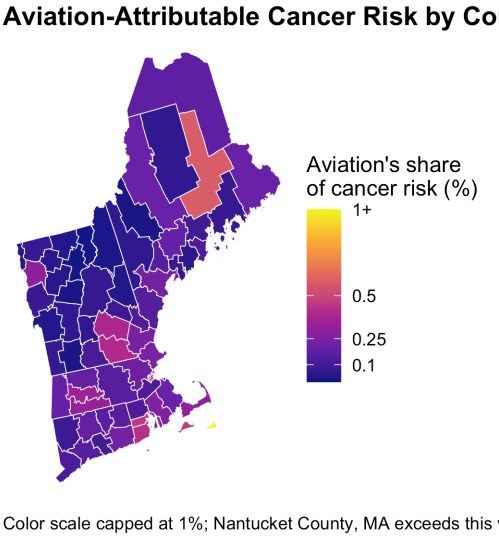


Figure 3: Modeled Aviation-Attributable Cancer Risk by County

The descriptive statistics and figures indicate aviation does have a measurable impact on modeled cancer risk in New England though exposure risk is heavily concentrated. The relationship between commercial enplanements and cancer risk attributable to airport pollution is roughly linear in log-log format, consistent with my expectations. However, there is a reasonable amount of variation, likely caused by factors not included in the data such

as aircraft type, untracked aircraft, and local factors such as proximity to residential areas. As expected by the massive advantage in aircraft movements, Massachusetts contains the counties with the highest cancer risk, and Maine counties with busy airports such as Bangor and Portland also show elevated levels. An interesting finding is Nantucket's position atop the rankings of highest percentage of cancer attributable to aviation. This is likely due to residents' close proximity to the busy, albeit seasonal, airport. Health risks are not distributed evenly across counties, as evidenced by the right skew of the histogram.

While these data support the goals of the research question, they have some shortcomings. First is the use of modeled estimates in place of observed outcomes. The "cancer risk" is not calculated by figuring out who developed cancer, but by a prediction of who will get it. Since the data is cross-sectional, it cannot show trends in the data. Though it made the data analysis process far simpler, the usage of county level data obscures the true impact of efforts to measure impacts on those who live closest to airports . Should more accurate health data become available, this would be the next step in my model creation. Finally, causal relationships will be difficult to establish due to a lack of control variables.

Discussion

The County-level analysis presented above produces a picture largely consistent with existing research but at a much smaller magnitude than targeted studies of major airports may suggest. This is not to say aviation-attributable cancer risk does not exist in New England. It is real, its association with airport activity is statistically significant, and concentrated in counties with high levels of aircraft activity, but rarely exceeds 0.5% of the cancer risk in any given county. Though this finding agrees with Hudda et al. and Riley et al. who found 2-5x increases in UFP concentrations near runways at major airports, the unit of measure is responsible for much of the apparent gap. These two papers measure exposure near large airports while AirToxScreen focuses on smaller airports and pollutant

data averaged to county-level. This means someone living 20 miles away from the airport is contained in the same average as someone whose home borders the airport, even though their exposures are dramatically different.

The Nantucket finding appears to be caused by its status as the county with the least dilution. Despite the lower traffic numbers of Nantucket's airport, the average location of year-round Nantucket residents is much closer to the airport than would be expected for other counties. However, the true cause of Nantucket's outlier status may be due to their lack of industrial and highway emissions, meaning a typical cancer risk from aviation makes up a much larger share of the overall risk (same numerator, smaller denominator).

Much of the data is not as accurate as one would have hoped for a project such as this. Future research into this topic would benefit from observed health outcomes instead of modeled ones. For instance, there is little public data on aviation-attributable risk for heart and respiratory disease. These cross-sectional samples of data suffice for looking at a point in time but are not as helpful for comparing trends over time such as those from the changes to the NextGen system in Newark. Another concern is a lack of robust controls. Areas with high-volume airports also have high levels of automotive traffic and industrial activity, both of which can produce similar emissions to turbine aircraft. While AirToxScreen attempts to separate the causes of pollution, it has proven difficult to get a precise estimate, especially for smaller General Aviation airports.

Conclusion

The current regulatory environment of aviation does not align with those impacted by exposure. Despite the ever decreasing cost of air travel, those who fly often and create regulation are not likely to experience most of its impact. The federal government is rarely held accountable, despite the fact that the FAA is responsible for aviation regulation in the United States and much of the worst chemical pollution comes from military airports. Com-

munities surrounding these airports carry the environmental scars of these legacy airports, and may not see a commensurate benefit from them today. Whether this is an environmental justice concern depends on your definition of choice, but the data have drawn some clear outcomes.

County-level analysis confirms a statistically significant association between aviation activity and modeled cancer risk across New England, though the magnitude is small. Aviation's share of cancer risk exceeds 0.5% in only one county, Nantucket. However, this does not mean the risk is negligible. County-level averaging, the highest resolution widely available data, dilutes the exposure of those who live close to airports and experience the highest costs. There are several policy implications resulting from these findings. First is the FAA's failure to certify a replacement to leaded aviation gasoline which continues to expose communities near general aviation airports to elevated levels of environmental lead. Second, EPA monitoring and emissions modeling should be expanded to smaller airports where data is less readily available. Finally, PFAS and related contaminant remediation should be implemented at both legacy and active military airports when applicable. None of these solutions require authority not already granted to the FAA or EPA. Future research would benefit from more granular data on observed health outcomes instead of modeled estimates, within-county resolution to capture a finer exposure gradient near airports, and panel data tracking changes in exposure over time. The data used here are sufficient to establish these relationships, but not adequate to measure the true effect in the most heavily impacted areas, those within just a few miles of the runway.

Appendix

Table 2: OLS Regression Results: Log Airport Cancer Risk on Log Enplanements

	Bivariate	With Airports	With Controls
Intercept	−4.384***	−4.416***	−7.257***
	(0.224)	(0.249)	(1.124)
Log Enplanements	0.138***	0.136***	0.116***
	(0.027)	(0.028)	(0.028)
Number of Airports		0.033	−0.045
		(0.104)	(0.103)
Log Population			0.261*
			(0.101)
R2	0.366	0.367	0.451
R2 Adj.	0.352	0.339	0.413
Num.Obs.	48	48	48

• $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

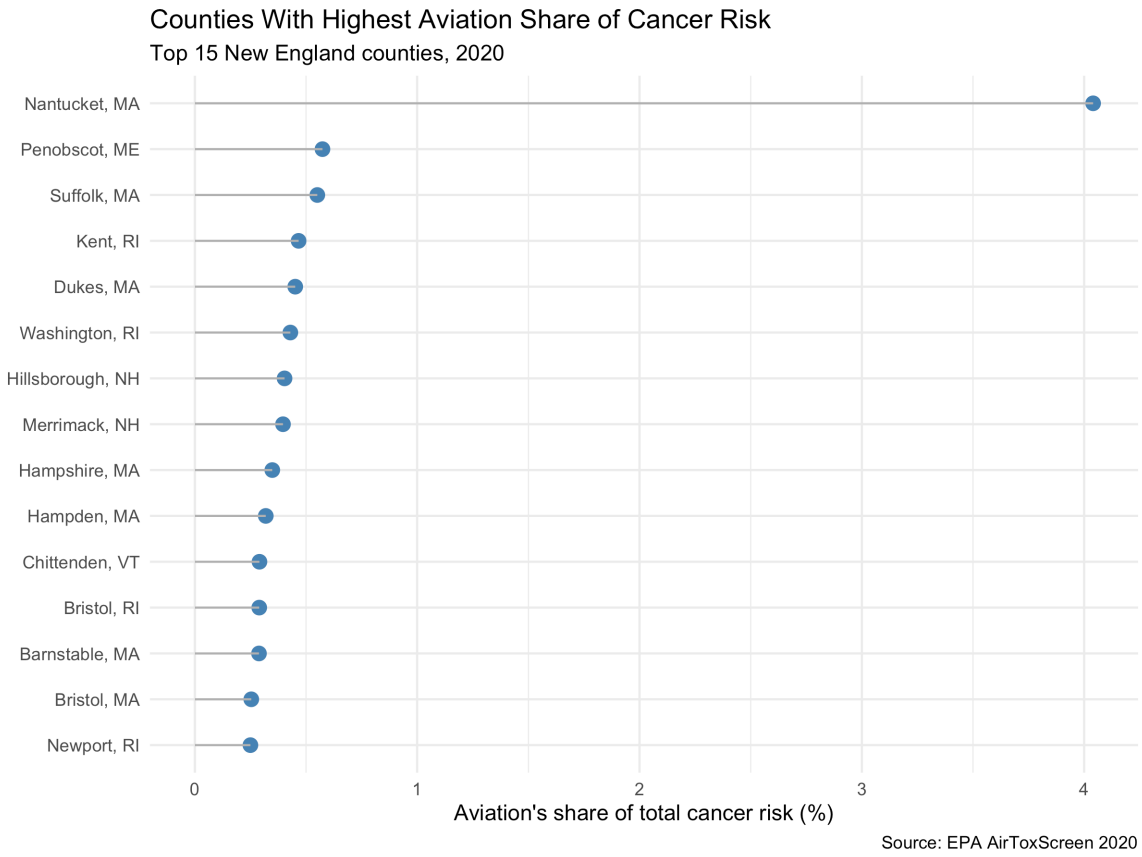


Figure 4: Counties With Highest Aviation Share of Cancer Risk

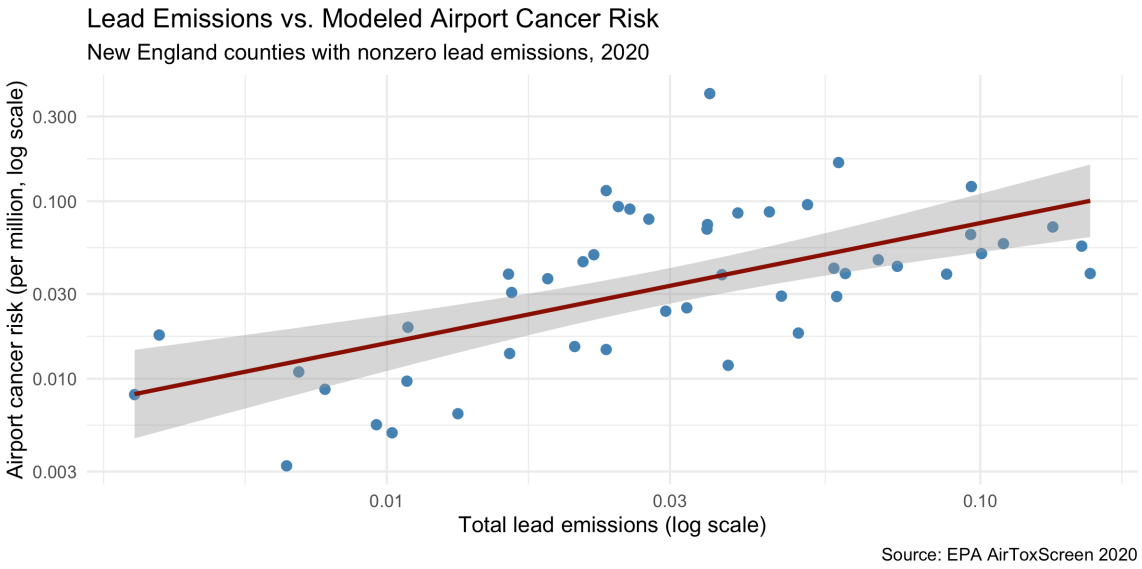


Figure 5: Lead Emissions vs. Modeled Airport Cancer Risk

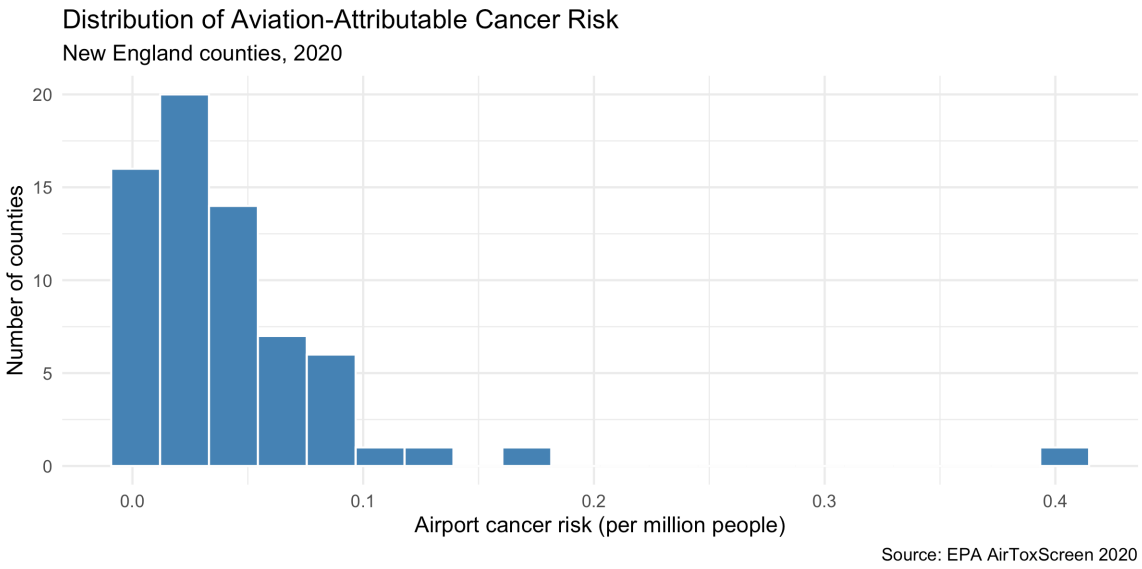


Figure 6: Distribution of Modeled Aviation-Attributable Cancer Risk in New England Counties

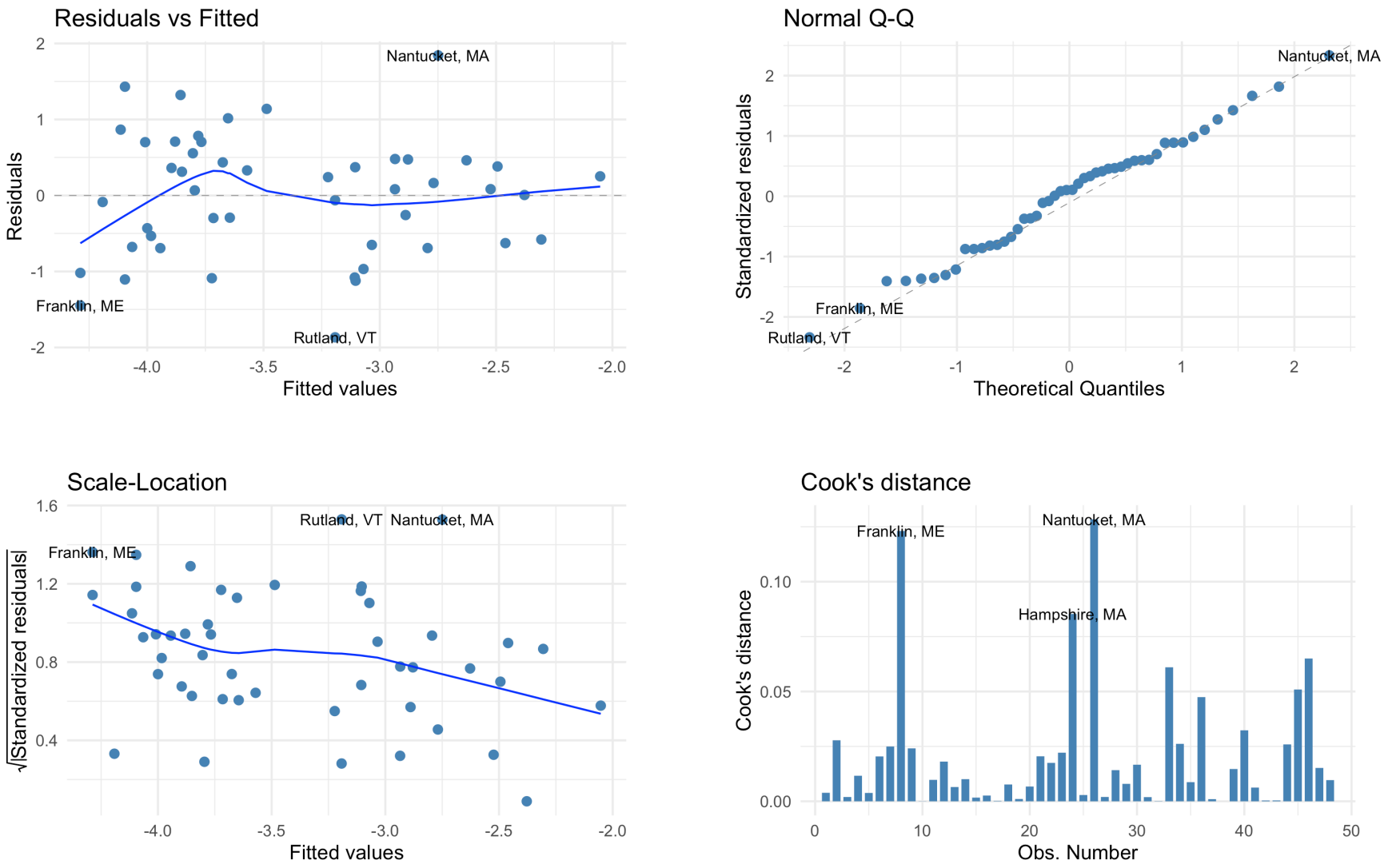


Figure 7: Regression Diagnostic Plots, Bivariate Log-Log Model

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